

# Deep-Research Prompt: AI for Philological Heritage

## — State of the Art & Best Cases

Paste this prompt into Claude (Research mode) or assign it to a dedicated research subagent. Replace bracketed placeholders before running.

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### Role

You are a senior digital-humanities strategist and web designer specialising in AI-assisted cultural-heritage production. You combine three literacies: (1) academic philology and medieval studies, (2) modern AI/LLM tooling, and (3) public-facing digital presentation (web, interactive, archival).

### Object of Study

I hold the collected articles of **[SCHOLAR NAME]** — a renowned historian, linguist, and expert in medieval literary works — as a **scanned/typeset PDF book** in **[LANGUAGE(S), e.g. Armenian, Russian, English, with quoted passages in Classical Armenian (Grabar), Greek, Latin]**. The corpus includes: textual criticism, manuscript studies, etymology, historical commentary, illustrations/plates of manuscripts, and a dense scholarly apparatus (footnotes, bibliography, indices).

### Mission

Produce a structured intelligence report (3,000–5,000 words + source list) covering the **latest developments (prioritise the last 24 months)** in AI applied to philological and medieval-heritage material, and the **best real-world cases** of analysing and presenting such heritage. The report will directly feed a technical specification (CLAUDE.md) for building a digital edition and public web presentation of this book.

### Research Axes

#### 1. Text acquisition from scholarly PDFs

- Current best OCR/HTR for multi-script, footnote-heavy academic books: Transkribus, eScriptorium/Kraken, Google Document AI, Azure Document Intelligence, Surya, and vision-LLM OCR (Claude, GPT-4V/o-series, Gemini) — accuracy on **[non-Latin script, e.g. Armenian]**, diacritics, mixed-language lines.

- Layout analysis: separating body text, footnotes, apparatus criticus, captions, running heads; recovering logical structure (article boundaries, headings).
- Post-correction workflows: LLM-assisted OCR correction, human-in-the-loop QA, measurable CER/WER benchmarks reported for comparable projects.

## 2. AI analysis of philological content

- Named-entity recognition for historical persons, places, manuscripts (shelfmarks), works, and dates in medieval-studies text; multilingual/ancient-language NER models.
- Citation and cross-reference extraction (biblical, classical, manuscript sigla).
- Semantic linking to authority files: VIAF, Wikidata, Pleiades, Pinakes, **[domain authorities, e.g. Matenadaran catalogues, Armenian Manuscript databases]**.
- LLM-based summarisation and abstracting of dense scholarly articles without distortion; approaches to hallucination control in scholarly contexts (RAG over the corpus itself, citation-grounded generation, refusal thresholds).
- Etymology/lexicography-specific AI work (relevant if the corpus includes linguistic articles): computational etymology, historical-lexicon digitisation projects.
- Stylometry and authorship/attribution tooling where relevant to medieval texts.

## 3. Standards and data models

- TEI-XML current practice for born-digital editions of collected scholarly papers (not just primary sources) — TEI Publisher, TEI Processing Model, CETELcean.
- IIIF (Image API + Presentation API v3) for plates and manuscript images; Mirador 4 and Universal Viewer state of play.
- Linked Open Data patterns for humanities (CIDOC-CRM, LIDO where museum objects appear); persistent identifiers (DOI, Handle, ARK) for article-level citation.

## 4. Retrieval and conversational access

- RAG architectures proven on humanities corpora: chunking strategies for footnoted scholarly prose, multilingual embeddings quality for **[LANGUAGE(S)]** (e.g. multilingual-e5, BGE-M3, Voyage, Cohere), hybrid BM25+dense retrieval.
- "Chat with the archive" deployments by libraries/museums — which ones are respected by the scholarly community, which drew criticism, and why.
- Question-answering with verifiable citation back to page/paragraph of the source PDF.

## 5. Presentation & web design best cases

Identify 8–12 exemplary projects and dissect each (institution, stack, AI components, design language, what to imitate, what to avoid). Prioritise:

- Digital scholarly editions and collected-works portals (e.g. recent TEI Publisher showcases, e-editiones community projects).

- Manuscript-heritage platforms: e.g. Vatican DigiVatLib, Fragmentarium, e-ktobe, Transkribus-derived portals, Cambridge Digital Library, **[regional exemplars, e.g. Matenadaran digital initiatives, Calouste Gulbenkian–funded Armenian projects]**.
- AI-enhanced storytelling in GLAM: Smithsonian, Europeana generative-AI pilots, Google Arts & Culture experiments — with critical assessment.
- Interactive visualisations: timelines (vis-timeline and successors), geographic layers, network graphs of persons/manuscripts, side-by-side facsimile + transcript.
- Typography and multilingual web design for the scripts involved; accessible design (WCAG 2.2) for long-form scholarly reading.

## 6. Ethics, rights, and scholarly acceptance

- Copyright status handling for a 20th–21st-century scholar's articles (estate permissions, publisher rights on offprints).
- Community norms on AI-generated summaries/translations of scholarship: disclosure, labelling, review requirements emerging in DH journals and library policy.
- Preservation: OAIS-aligned archiving, static-site longevity vs. dynamic apps.

## Method Requirements

- Search in English **and** in **[other relevant languages]**; scholarly sources (DH journals, conference proceedings — DH, EADH, CHR, ICDAR, IIF community) take precedence over vendor marketing.
- Date-stamp every claim; flag anything older than 2023 as "established practice" vs. newer items as "emerging".
- For each best case, capture at least one screenshot-worthy URL for a design-reference moodboard.
- Where evidence conflicts (e.g. OCR benchmark claims), present both sides and state your confidence.

## Deliverable Format

1. **Executive summary** (max 400 words) — the five decisions this research settles.
2. **Findings by axis** (sections 1–6 above), each ending with "Implications for our build" (3–5 bullets max).
3. **Best-case gallery** — table: project | institution | AI used | stack | steal-worthy ideas | pitfalls.
4. **Recommended reference architecture** — one paragraph + component list naming concrete tools/models, ready to be transcribed into a CLAUDE.md specification.
5. **Source list** with dates and one-line relevance notes.